

Agent Trust Fabric

Cryptographic Authority Governance for Autonomous AI Agents



OMNIX QUANTUM

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Author: Harold Nunes

Institution: OMNIX QUANTUM LTD

Contact: harold@omnixquantum.com

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Abstract

Autonomous AI agents increasingly make consequential decisions in enterprise systems — executing trades, authorizing medical procedures, routing logistics, and managing critical infrastructure. Yet the governance of *who authorized these agents, under what authority, and whether that authority was valid at the moment of execution* remains unaddressed by existing frameworks.

This paper presents the OMNIX Agent Trust Fabric (ATF) — a formally specified, post-quantum cryptographic protocol that answers four questions for every AI agent execution: (1) Who authorized this agent? — via ML-DSA-65 signed Delegation Receipt; (2) What authority did it hold? — via Monotonic Authority Reduction, TLA+ verified; (3) Was the authority valid at execution? — via Temporal Admissibility Record captured before the pipeline runs; (4) Is the full chain independently verifiable? — via offline CLI verifier, no platform access required.

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1. Problem Statement

1.1 The AI Agent Authorization Gap

Modern AI governance frameworks address *what decisions are made* but not *who authorized the agent that made them*. When an enterprise AI agent executes a high-stakes action, the audit record typically captures:

- What decision was made
- What signals influenced the decision
- A cryptographic hash of the decision receipt

What is systematically missing:

- **Proof that the agent was authorized to act** — not just authenticated, but explicitly authorized with bounded scope
- **Proof that the authority was valid at the nanosecond of execution** — not valid yesterday, not valid in principle, but valid *then*
- **A verifiable chain from the AI agent back to a human principal** — traceable without trusting the platform

This gap is not a limitation of existing governance frameworks — it is not their design target. OAuth 2.0 handles access delegation; W3C Verifiable Credentials handle identity claims; JWT handles session validity. None was designed to answer: *"Was this autonomous AI agent's authority mathematically bounded and verified at the moment it executed this specific governance decision?"*

1.2 The Accountability Requirement

Regulation is closing this gap:

Regulation	Requirement	ATF Response
EU AI Act Art. 13	Transparency for high-risk AI — "who authorized this system"	ATF DR chain to Tier-1 human
EU AI Act Art. 14	Human oversight — human principal must be identifiable	ATF chain_root_id → Tier-1
DORA Art. 9	ICT resilience — documented controls for automated systems	ATF TAR proves control at execution
MiCA Rec. 65	Algorithmic trading controls — authorization audit trail	ATF triple chain for trading decisions
NIST AI RMF GOVERN	AI decisions traceable to accountable entities	ATF cryptographic traceability
SOC 2 CC6	Logical access controls — non-repudiation	ATF PQC signatures

1.3 Why Existing Protocols Are Insufficient

Protocol	What It Provides	What It Cannot Provide
OAuth 2.0	Access token delegation	Monotonic authority bound; pre-execution temporal record; human chain traceability
JWT (RFC 7519)	Signed claims with expiry	Authority budget arithmetic; separate admissibility artifact; offline chain verification
W3C VC	Signed identity claims	Authority bounds; temporal admissibility; mandatory cross-domain reduction
SPIFFE/SPIRE	Workload identity	Authority delegation chain; governance receipt integration
OpenID Connect	Authentication	Authorization chain governance; PQC signatures; formal invariants

None of these protocols is wrong — they solve different problems. ATF addresses the gap that exists at the intersection of all of them: **authority governance for autonomous AI agents, before and during execution.**

2. The ATF Protocol Architecture

2.1 Core Concepts

Agent Identity Record (AIR)

Every AI agent registered in an ATF-compliant system receives a globally unique identifier: `AID-{DOMAIN}-{16HEX}`. The AIR contains the agent's authority budget cap, domain, registration tier, and public key for verification.

Delegation Receipt (DR / ATFDR-{16HEX})

A PQC-signed record issued by a principal (human or agent) to an agent, explicitly granting a bounded subset of the principal's authority for a defined task scope. Content-hashed (SHA-256 over canonical JSON), signed with ML-DSA-65 (Dilithium-3, NIST FIPS 204), and independently verifiable offline using the embedded delegator public key.

Temporal Admissibility Record (TAR / ATFTAR-{16HEX})

A PQC-signed record issued at the exact moment of admission to the governance pipeline — **before any governance logic executes**. The TAR proves: (1) the DR was ACTIVE at `execution_ns` (nanosecond-resolution); (2) the admission decision was ADMITTED or REJECTED; (3) the `execution_ref` binds this TAR to a specific GovernanceReceipt.

Trust Lattice

The directed acyclic graph (DAG) of all delegation receipts. The lattice enforces the Acyclicity invariant (ATF-INV-002) and provides chain traversal for complete authority verification.

GovernanceReceipt

The governance decision record (ADR-028), extended with `atf_context` embedding the DR, TAR, and trust summary — forming the three-artifact audit chain.

2.2 The Authority Lifecycle

Phase 1 – Registration

Human Tier-1 registers agent → AIR created → authority budget assigned

Phase 2 – Delegation

Principal signs DR → `authority_budget_granted ≤ authority_budget_delegator`

DR stored in trust lattice → `chain_root_id` set

Phase 3 – Admission

Agent presents DR at execution boundary

`ATFConnector.admit()` called BEFORE governance pipeline

```

TAR issued: execution_ns captured, DR status verified, TAR signed
TAR.execution_ref = GovernanceReceipt.receipt_id

Phase 4 – Execution
Governance pipeline runs (AVM, AI, veto chain)
Decision produced with PQC-signed GovernanceReceipt
ATF context embedded: delegation_id, tar_id, authority_budget

Phase 5 – Verification
Any party runs: python omnix_atf_verify.py receipt.json
DR signature verified against embedded public key
TAR status confirmed ADMITTED
execution_ref cross-checked with GovernanceReceipt.receipt_id
Chain root traced to Tier-1 human principal
Result: VERIFIED or NOT VERIFIED

```

2.3 Authority Budget Arithmetic

The authority budget is a real number in $[0.0, 100.0]$ representing the fraction of full authority held by an agent. The Monotonic Authority Reduction (MAR) invariant requires:

```

For all DRs D in the trust lattice:
D.authority_budget_granted ≤ D.authority_budget_delegator

For all chains C = [DR1, DR2, ..., DRn]:
DR1.authority_budget_granted ≥ DR2.authority_budget_granted ≥ ... ≥
DRn.authority_budget_granted

```

This invariant is formally specified in TLA+ and verified by model checking over bounded state spaces (ATF-FV-1.0). It ensures that delegation cannot *create* authority — only distribute and bound it.

Example delegation chain:

```

HUMAN-TIER1 budget = 100.0 (full authority)
■■ ATFDR-ROOT → AID-FINANCE-001 budget = 80.0 (20% reduction)
■■ ATFDR-A → AID-FINANCE-002 budget = 50.0 (37.5% reduction)
■■ ATFDR-B → AID-FINANCE-003 budget = 20.0 (60% reduction)

```

3. Formal Invariants

ATF-1.0 defines six formally specified invariants. All are mandatory for ATF-COMPLIANT-LEVEL-2 and above.

Invariant	Identifier	Formal Statement
Monotonic Authority Reduction	ATF-INV-001	$DR.budget_granted \leq DR.budget_delegator$ for all DRs
Acyclicity	ATF-INV-002	The trust lattice is a DAG — no delegation cycle exists
Chain Root Consistency	ATF-INV-003	All DRs in a chain share the same chain_root_id
Content Hash Immutability	ATF-INV-004	DR fields are immutable post-issuance (hash binds all fields)
Temporal Non-Future-Dating	ATF-INV-005	$TAR.execution_ns \leq \text{current time at verification}$
Independent Verifiability	ATF-INV-006	Full chain verifiable offline using only receipts and root public key

TLA+ Coverage: INV-001 (MARInvariant), INV-002 (AcyclicityInvariant), INV-003 (ChainRootConsistency), and INV-004 (ImmutabilityProperty) are formally specified and model-checked in docs/formal/ATF-TLA-SPEC.tla. This is the same formal verification methodology used by Amazon Web Services for DynamoDB.

4. Cryptographic Specification

4.1 Signature Algorithm

Property	Value
Algorithm	ML-DSA-65 (Dilithium-3)
Standard	NIST FIPS 204 (August 2024)
Security level	NIST PQC Level 3
Public key size	1,952 bytes
Signature size	3,293 bytes
Hash function	SHA-256 (content hash construction)
Encoding	Base64 standard (signature), hex (content hash)

4.2 Content Hash Construction

All ATF artifacts use a deterministic content hash for signature binding:

```
canonical = json.dumps(fields, sort_keys=True, separators=(',', ':'), ensure_ascii=True)
content_hash = hashlib.sha256(canonical.encode('utf-8')).hexdigest()
```

Artifact	Hashed Fields
DR	delegation_id, delegator_id, delegate_id, task_scope, authority_budget_granted, authority_budget_delegator, chain_root_id, delegation_depth, expires_at, status, created_at
TAR	tar_id, delegation_id, agent_id, execution_ref, execution_ns, dr_status_at_admission, dr_expires_at, authority_budget, domain, task_action, admission_status, chain_root_id
DTR	dtr_id, source_delegation_id, source_domain, target_domain, source_agent_id, target_agent_id, source_authority_budget, translated_budget, translation_discount, chain_root_id

4.3 Fallback Mode

When ML-DSA-65 is unavailable, ATF falls back to content hash only (no PQC signature). This mode (`pqc_signature = null`) is explicitly: **Permitted** for ATF-COMPLIANT-LEVEL-1 development/test environments; **Prohibited** for ATF-COMPLIANT-LEVEL-2+ production deployments.

5. Temporal Admissibility (ADR-157)

The Temporal Admissibility Record (TAR) is ATF's mechanism for proving that an agent's authority was valid at the exact moment of execution.

5.1 Why TAR Is Necessary

JWT exp claims verify that a token is not expired at the time of checking. This is an implicit check — there is no artifact proving the check occurred. TAR makes the check explicit and produces a signed, persistent record:

Property	JWT exp check	TAR (ATF-INV-005)
Existence check	Yes	Yes
Signed artifact	No	Yes (ML-DSA-65)
Timestamp granularity	Second	Nanosecond-resolution
Persistence	No	Yes (database row, immutable)
Binding to execution	No	Yes (execution_ref)
Independent verifiability	No	Yes (offline CLI)
Rejection record	No	Yes (REJECTED status with reason)

5.2 TAR Issuance Protocol

```
Input: DR, agent_id, task_action, execution_ref
Output: TemporalAdmissibilityRecord (ATFTAR-{16HEX})

1. execution_ns ← time.time_ns() # Nanosecond capture – BEFORE checks
2. Verify DR.status == ACTIVE
3. Verify execution_ts < DR.expires_at
4. Verify execution_ts >= DR.created_at
5. Set admission_status ← ADMITTED (or REJECTED with reason)
6. Compute content_hash over all TAR fields including execution_ns
7. Sign content_hash with ML-DSA-65
8. Persist TAR to atf_temporal_records table
9. Return TAR
```

Step 1 is executed before steps 2–4. This ensures the execution_ns reflects the actual admission time, not the outcome of the checks. The TAR documents the authority state *at admission*, before any decision logic executes.

6. Cross-Domain Trust Portability (ADR-158)

In multi-domain deployments, an agent authorized in domain FINANCE may need to access resources in domain HEALTHCARE. Cross-domain trust requires explicit authority translation with mandatory reduction.

6.1 Domain Translation Receipt (DTR / ATFDTR-{{16HEX}})

```
source_budget = 60.0 (FINANCE domain)
discount = 0.30 (FINANCE → HEALTHCARE policy)
translated_budget = 60.0 × (1 - 0.30) = 42.0 (HEALTHCARE domain)
```

The discount is mandatory — cross-domain translation **MUST** reduce authority (CDTP-INV-003). An agent cannot gain authority by crossing domains.

6.2 Standard Domain-Pair Discount Policies

Source → Target	Default Discount	Rationale
HEALTHCARE → FINANCE	30%	High sensitivity of both domains
FINANCE → HEALTHCARE	30%	Symmetric high-sensitivity
FINANCE → INSURANCE	15%	Adjacent regulatory domain
DEFENSE → FINANCE	40%	Strict isolation requirement
DEFENSE → HEALTHCARE	45%	Maximum cross-domain sensitivity
Other pairs	20%	Default cross-domain discount

7. Governance Receipt Integration

7.1 The Triple Chain

Every OMNIX governance decision originating from an ATF-registered agent produces three cross-referenced artifacts:

```
{
  "receipt_id": "OMNIX-FIN-20260512-A3F7B2",
  "decision": "APPROVED",
  "pqc_signature": "ML-DSA-65 signature ...",
  "atf_context": {
    "delegation_id": "ATFDR-8B2C4D6E1F3A5B7C",
    "tar_id": "ATFTAR-C4D8E2F1A3B5C7D9",
    "agent_id": "AID-FINANCE-3A7F9B2C1D4E5F6A",
    "delegator_id": "HUMAN-TIER1-HN-001",
    "admission_status": "ADMITTED",
    "execution_ns": 1747058400000000000,
    "authority_budget": 60.0,
    "chain_root_id": "ATFDR-8B2C4D6E1F3A5B7C",
    "pqc_signed": true
  }
}
```

This structure enables a verifier to answer all four questions using only this JSON document and the root public key.

7.2 Verification Flow

```
verify(receipt.json):
1. Check receipt.pqc_signature over receipt.content_hash → receipt integrity
2. Fetch DR by atf_context.delegation_id → authority claim
3. Verify DR.pqc_signature over DR.content_hash → delegation integrity
4. Verify DR.delegator_public_key signed the hash → principal identity
5. Fetch TAR by atf_context.tar_id → temporal proof
6. Verify TAR.pqc_signature over TAR.content_hash → TAR integrity
7. Check TAR.admission_status == ADMITTED → execution admitted
8. Check TAR.execution_ref == receipt.receipt_id → binding holds
9. Traverse chain to chain_root_id → human traceability
Result: VERIFIED | NOT VERIFIED

All steps can be performed offline using only the receipt artifacts.
```

8. Independent Verifiability (ATF-INV-006)

"Any party MUST be able to verify a delegation chain using only the receipts and the root public key. No access to the issuing platform, API, account, or internet connection is required."

This property distinguishes ATF from platform-dependent audit trails. A regulator conducting a post-incident review can:

- Download the receipt artifacts from the institution's disclosure
- Run `python omnix_atf_verify.py receipt.json`
- Receive a complete VERIFIED/NOT VERIFIED verdict with full chain analysis
- Without any interaction with OMNIX or the deploying institution's systems

This is analogous to how anyone can verify a PGP-signed document without contacting the key issuer — but applied to the entire AI agent authority chain.

9. Compliance Framework

ATF defines three compliance levels:

Level 1 — Basic (Development and Proof-of-Concept)

- DR content hash (SHA-256)
- MAR enforcement
- Chain traversal
- PQC signature: SHOULD (not required)
- TAR: OPTIONAL

Level 2 — Standard (Production, Enterprise)

- All Level-1 requirements
- PQC signature REQUIRED (ML-DSA-65)
- TAR REQUIRED for all governance-connected executions
- ATF CCS ≥ 80 required for high-assurance domains
- Independent verifiability demonstrated

Level 3 — Sovereign (Regulated, Critical Infrastructure)

- All Level-2 requirements
- RFC 3161 TSA timestamp counter-signature on TARs
- FIPS 140-3 validated cryptographic module SHOULD be used
- Formal verification documentation (TLA+ or equivalent)
- Public CLI verifier available for independent audit
- Cascade revocation within 1 hour of compromise detection

10. Interoperability

Framework / Protocol	Relationship to ATF
W3C Verifiable Credentials	ATF DRs share structural concepts. ATF extends with authority budget arithmetic, MAR invariant, TAR as separate signed artifact, and TLA+-verified formal invariants.
IETF JWT (RFC 7519)	ATF receipts can be serialized as JWT claims. The exp claim $\approx$ DR.expires_at. TAR provides stronger evidence than exp — signed artifact, not just a claim.
OpenID Connect	ATF agent identities can be registered as OAuth 2.0 clients. chain_root_id can be used as sub claim linking to human principal's OIDC identity.
SPIFFE/SPIRE	ATF extends workload identity with delegation chain, authority arithmetic, and governance receipt integration.

Framework / Protocol	Relationship to ATF
LangChain / AutoGen / CrewAI	ATF is framework-agnostic. ATFConnector integrates into any Python-based agent framework without modifying existing logic.

11. Implementation Reference

Module	Purpose
trust_lattice.py	TrustLattice, AgentIdentity, DelegationEngine — DAG management and MAR enforcement
temporal_authority.py	TemporalAuthorityEngine, TemporalAdmissibilityRecord — TAR issuance and verification
domain_bridge.py	CrossDomainBridge, DomainTranslationReceipt — cross-domain authority translation
atf_connector.py	ATFConnector, ATFContext — integration bridge for governance pipeline
agent_blueprint.py	REST API endpoints: /api/atf/register, /api/atf/delegate, /api/atf/admit, /api/atf/verify
omnix_atf_verify.py	Standalone offline CLI verifier — zero dependencies beyond Python 3.8+
ATF-TLA-SPEC.tla	TLA+ formal specification — 5 invariants model-checked
test_agent_trust_fabric.py	50+ protocol tests covering all invariants and error conditions

12. Known Limitations and Open Questions

Limitation	Mitigation	Status
Clock trust in TAR (TM-004)	RFC 3161 TSA integration (planned Level-3 requirement)	OPEN
Revocation propagation latency (TM-006)	Short DR validity + centralized revocation registry	PARTIAL
Theorem-proved MAR vs model-checked	Port to Coq/Lean for one invariant (planned)	PLANNED
FIPS 140-3 validated library	Library-agnostic spec; swap pqc for validated lib	OPEN
Multi-instance TAR uniqueness	Redis SETNX / DB unique constraint	PARTIAL

13. Related Work

W3C Verifiable Credentials: Identity claims with PQC extensions (emerging drafts). No authority budgets or temporal admissibility artifacts.

SPIFFE/SPIRE: Workload identity for service meshes. No delegation chains or authority arithmetic.

OpenID Connect: Authentication protocol. No authorization governance chain.

NIST SP 800-207 (Zero Trust): Zero Trust Architecture principles. ATF implements explicit trust verification before each execution event.

NIST AI RMF (January 2023): AI Risk Management Framework. ATF directly implements accountability requirements in the GOVERN function.

EU AI Act (August 2024): ATF addresses Arts. 13/14 (transparency and human oversight) for high-risk AI systems.

CRYSTALS-Dilithium / ML-DSA: NIST FIPS 204 (August 2024). ATF adopts ML-DSA-65 as mandatory signature algorithm for Level-2+ compliance.

14. Conclusion

The OMNIX Agent Trust Fabric addresses a specific, well-defined gap in AI governance infrastructure: the absence of cryptographic proof of *who authorized an AI agent, under what authority bound, when that authority was valid, and whether execution was admitted at that exact moment*.

ATF provides this through a three-artifact chain (DR + TAR + GovernanceReceipt), formally specified invariants (TLA+ model-checked), post-quantum cryptography (ML-DSA-65, NIST FIPS 204), and independent offline verifiability (ATF-INV-006).

The design is complementary to existing orchestration frameworks, not a replacement for them. It operates at the governance layer that precedes execution — making the authority chain verifiable before, during, and after an AI agent acts.

As autonomous AI systems take on greater responsibility in enterprise and regulated environments, this governance layer becomes a prerequisite for institutional trust. OMNIX ATF provides the formal specification, reference implementation, and public verification tooling to establish that trust on a cryptographic foundation.

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